

API Reference - Horizon Medical

Versión: 2.0

Base URL: `https://api.horizon-medical.com/v2`

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Autenticación

Obtener Token JWT

Endpoint: `POST /auth/login`

```
curl -X POST https://api.horizon-medical.com/v2/auth/login \
-H "Content-Type: application/json" \
-d '{
  "email": "doctor@hospital.com",
  "password": "securepassword123",
  "mfaToken": "123456" # opcional si MFA está habilitado
}'
```

Response (200 OK):

```
{
  "success": true,
  "data": {
    "accessToken": "eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...",
    "refreshToken": "eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...",
    "expiresIn": 3600,
    "user": {
      "id": "user_507f1f77bcf86cd799439011",
      "email": "doctor@hospital.com",
      "name": "Dr. Juan López",
      "role": "doctor",
      "specialization": "cardiología"
    }
  }
}
```

Renovar Token

Endpoint: POST /auth/refresh

```
curl -X POST https://api.horizon-medical.com/v2/auth/refresh \
  -H "Authorization: Bearer {refreshToken}" \
  -H "Content-Type: application/json"
```

Response (200 OK):

```
{
  "success": true,
  "data": {
    "accessToken": "eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...",
    "expiresIn": 3600
  }
}
```

Headers de Autenticación

```
Authorization: Bearer {accessToken}
Content-Type: application/json
X-API-Key: optional_api_key # Para integraciones
```

Estructura de Respuestas

Respuesta Exitosa

```
{
  "success": true,
  "data": {
    // Objeto o array con datos
  },
  "meta": {
    "timestamp": "2026-01-15T14:30:00Z",
    "requestId": "req_507f1f77bcf86cd799439011",
    "version": "2.0"
  }
}
```

Respuesta con Paginación

```
{
  "success": true,
  "data": [
    // array de items
  ],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 150,
    "totalPages": 8,
    "hasNextPage": true,
    "hasPreviousPage": false
  }
}
```

Respuesta de Error

```
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Validation failed",
    "details": [
      {
        "field": "email",
        "message": "Invalid email format"
      }
    ]
  },
  "meta": {
    "timestamp": "2026-01-15T14:30:00Z",
    "requestId": "req_507f1f77bcf86cd799439011"
  }
}
```

Códigos de Estado HTTP

Código	Significado	Caso de Uso
200	OK	Solicitud exitosa
201	Created	Recurso creado exitosamente
202	Accepted	Procesando (jobs asíncronos)
204	No Content	Éxito sin contenido de respuesta
400	Bad Request	Parámetros inválidos
401	Unauthorized	Autenticación requerida/inválida
403	Forbidden	Autorización insuficiente
404	Not Found	Recurso no encontrado
409	Conflict	Conflicto de datos
422	Unprocessable Entity	Validación fallida
429	Too Many Requests	Rate limit excedido
500	Internal Server Error	Error del servidor
503	Service Unavailable	Servicio en mantenimiento

Rate Limiting

Límites por Tier

Free Tier:

- └ 100 requests/hour
- └ 1000 requests/day
- └ Max 10 concurrent

Professional Tier:

- └ 10,000 requests/hour
- └ 100,000 requests/day
- └ Max 100 concurrent

Enterprise Tier:

- └ Unlimited
- └ Unlimited
- └ Custom limits

Headers de Rate Limiting

```
X-RateLimit-Limit: 10000
X-RateLimit-Remaining: 9950
X-RateLimit-Reset: 1705348800
```

Manejo de Límite Excedido

HTTP 429 **Too Many Requests**

```
{
  "success": false,
  "error": {
    "code": "RATE_LIMIT_EXCEEDED",
    "message": "Too many requests. Please retry after 60 seconds.",
    "retryAfter": 60
  }
}
```

Usuarios

Obtener Perfil del Usuario Actual

Endpoint: GET /users/me

```
curl -X GET https://api.horizon-medical.com/v2/users/me \
-H "Authorization: Bearer {accessToken}"
```

Response (200 OK):

```
{
  "success": true,
  "data": {
    "id": "user_507f1f77bcf86cd799439011",
    "email": "doctor@hospital.com",
    "firstName": "Juan",
    "lastName": "López",
    "role": "doctor",
    "specialization": "cardiología",
    "institution": "Hospital Central",
    "mfaEnabled": true,
    "createdAt": "2025-06-15T10:30:00Z",
    "lastLogin": "2026-01-15T14:30:00Z"
  }
}
```

Actualizar Perfil

Endpoint: PATCH /users/me

```
curl -X PATCH https://api.horizon-medical.com/v2/users/me \
-H "Authorization: Bearer {accessToken}" \
-H "Content-Type: application/json" \
-d '{
  "firstName": "Juan",
  "lastName": "López García",
  "specialization": "cardiología_pediátrica"
}'
```

Cambiar Contraseña

Endpoint: POST /users/me/change-password

```
curl -X POST https://api.horizon-medical.com/v2/users/me/change-password \
-H "Authorization: Bearer {accessToken}" \
-H "Content-Type: application/json" \
-d '{
  "currentPassword": "oldpass123",
  "newPassword": "newpass456",
  "confirmPassword": "newpass456"
}'
```

Pacientes

Listar Pacientes

Endpoint: GET /patients

```
curl -X GET "https://api.horizon-medical.com/v2/patients?page=1&limit=20&search=juan" \
-H "Authorization: Bearer {accessToken}"
```

Parámetros Query:

```
page: number (default: 1)
limit: number (default: 20, max: 100)
search: string (por nombre, email, ID)
status: string (active, inactive, archived)
sortBy: string (name, createdAt, lastMonitoring)
sortOrder: string (asc, desc)
```

Response (200 OK):

```
{
  "success": true,
  "data": [
    {
      "id": "patient_507f1f77bcf86cd799439011",
      "firstName": "Juan",
      "lastName": "García",
      "email": "juan@example.com",
      "dateOfBirth": "1965-03-20",
      "gender": "M",
      "medicalRecord": "MR-12345",
      "status": "active",
      "lastMonitoring": "2026-01-15T10:00:00Z",
      "activeDevices": 1,
      "totalSessions": 24,
      "primaryPhysician": "Dr. López"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 150,
    "totalPages": 8
  }
}
```

Crear Paciente

Endpoint: POST /patients

```
curl -X POST https://api.horizon-medical.com/v2/patients \
-H "Authorization: Bearer {accessToken}" \
-H "Content-Type: application/json" \
-d '{
  "firstName": "Juan",
  "lastName": "García",
  "email": "juan@example.com",
  "phone": "+34612345678",
  "dateOfBirth": "1965-03-20",
  "gender": "M",
  "medicalHistory": "Hipertensión, diabetes tipo 2",
  "medicamentos": ["Lisinopril 10mg", "Metformina 500mg"],
  "alergias": "Penicilina",
  "emergencyContact": "María García (hermana) +34612345679"
}'
```

Obtener Paciente

Endpoint: GET /patients/{patientId}

```
curl -X GET https://api.horizon-medical.com/v2/patients/patient_507f1f77bcf86cd799439011 \
-H "Authorization: Bearer {accessToken}"
```

Actualizar Paciente

Endpoint: PATCH /patients/{patientId}

```
curl -X PATCH https://api.horizon-medical.com/v2/patients/patient_507f1f77bcf86cd799439011 \
-H "Authorization: Bearer {accessToken}" \
-H "Content-Type: application/json" \
-d '{
  "medicalHistory": "Hipertensión, diabetes tipo 2, nuevo diagnóstico",
  "medicamentos": ["Lisinopril 10mg", "Metformina 1000mg", "Atorvastatina 20mg"]
}'
```

Dispositivos Holter

Listar Dispositivos

Endpoint: GET /devices

```
curl -X GET "https://api.horizon-medical.com/v2/devices?status=active&type=holter" \
-H "Authorization: Bearer {accessToken}"
```

Response:

```
{
  "success": true,
  "data": [
    {
      "id": "device_507f1f77bcf86cd799439011",
      "serialNumber": "HM-001-5A2C",
      "status": "active",
      "type": "holter",
      "modelVersion": "2.0",
      "batteryLevel": 87,
      "lastSyncTime": "2026-01-15T14:30:00Z",
      "assignedPatient": "patient_507f1f77bcf86cd799439012",
      "signalQuality": 95,
      "firmwareVersion": "3.2.1",
      "calibrationDate": "2025-12-20T10:00:00Z"
    }
  ]
}
```

Asignar Dispositivo a Paciente

Endpoint: POST /devices/{deviceId}/assign

```
curl -X POST https://api.horizon-medical.com/v2/devices/device_507f1f77bcf86cd799439011/assign \
-H "Authorization: Bearer {accessToken}" \
-H "Content-Type: application/json" \
-d '{
  "patientId": "patient_507f1f77bcf86cd799439012",
  "monitoringDuration": 48,
  "syncFrequency": 30,
  "alertsEnabled": true
}'
```


Obtener Estado del Dispositivo

Endpoint: GET /devices/{deviceId}/status

```
curl -X GET https://api.horizon-medical.com/v2/devices/
device_507f1f77bcf86cd799439011/status \
-H "Authorization: Bearer {accessToken}"
```

Response:

```
{
  "success": true,
  "data": {
    "batteryLevel": 87,
    "batteryStatus": "charging",
    "lastSync": "2026-01-15T14:30:00Z",
    "signalQuality": 95,
    "isConnected": true,
    "lastLocation": {"latitude": 40.4168, "longitude": -3.7038},
    "storageFull": false,
    "errors": []
  }
}
```

Sesiones ECG

Iniciar Sesión de Monitoreo

Endpoint: POST /sessions

```
curl -X POST https://api.horizon-medical.com/v2/sessions \
-H "Authorization: Bearer {accessToken}" \
-H "Content-Type: application/json" \
-d '{
  "patientId": "patient_507f1f77bcf86cd799439012",
  "deviceId": "device_507f1f77bcf86cd799439011",
  "durationHours": 48,
  "syncFrequencySeconds": 30,
  "notes": "Evaluación de síncope"
}'
```

Response (201 Created):

```
{
  "success": true,
  "data": {
    "id": "session_507f1f77bcf86cd799439013",
    "patientId": "patient_507f1f77bcf86cd799439012",
    "deviceId": "device_507f1f77bcf86cd799439011",
    "status": "active",
    "startTime": "2026-01-15T14:30:00Z",
    "estimatedEndTime": "2026-01-17T14:30:00Z",
    "dataPoints": 0,
    "qualityScore": null
  }
}
```

Obtener Sesión

Endpoint: GET /sessions/{sessionId}

```
curl -X GET https://api.horizon-medical.com/v2/sessions/session_507f1f77bcf86cd799439013 \
-H "Authorization: Bearer {accessToken}"
```

Listar Sesiones de Paciente

Endpoint: GET /patients/{patientId}/sessions

```
curl -X GET "https://api.horizon-medical.com/v2/patients/patient_507f1f77bcf86cd799439012/sessions?page=1&limit=10" \
-H "Authorization: Bearer {accessToken}"
```

Registrar Evento en Sesión

Endpoint: POST /sessions/{sessionId}/events

```
curl -X POST https://api.horizon-medical.com/v2/sessions/session_507f1f77bcf86cd799439013/events \
-H "Authorization: Bearer {accessToken}" \
-H "Content-Type: application/json" \
-d '{
  "type": "symptoms",
  "category": "palpitations",
  "timestamp": "2026-01-15T16:45:00Z",
  "description": "Palpitaciones notables durante actividad",
  "severity": "moderate",
  "symptoms": ["palpitations", "shortness_of_breath"]
}'
```

Obtener Datos ECG de Sesión

Endpoint: GET /sessions/{sessionId}/ecg-data

```
curl -X GET "https://api.horizon-medical.com/v2/sessions/session_507f1f77bcf86cd799439013/ecg-data?derivation=II&limit=1000&offset=0" \
-H "Authorization: Bearer {accessToken}"
```

Parámetros:

```

derivation: string (I, II, III, aVR, aVL, aVF, V1-V6)
limit: number (max: 10000)
offset: number (para paginación)
format: string (json, csv, binary)

```

Response:

```

{
  "success": true,
  "data": {
    "sessionId": "session_507f1f77bcf86cd799439013",
    "derivation": "II",
    "samplingRate": 500,
    "unit": "mV",
    "samples": [
      {"timestamp": "2026-01-15T14:30:00.000Z", "value": 0.5},
      {"timestamp": "2026-01-15T14:30:00.002Z", "value": 0.6},
      // ... más samples
    ]
  },
  "pagination": {
    "total": 144000000,
    "offset": 0,
    "limit": 1000
  }
}

```

Análisis y Reportes

Generar Reporte

Endpoint: POST /sessions/{sessionId}/report

```

curl -X POST https://api.horizon-medical.com/v2/sessions/session_507f1f77bcf86cd799439013/report \
-H "Authorization: Bearer {accessToken}" \
-H "Content-Type: application/json" \
-d '{
  "reportType": "comprehensive",
  "format": "pdf",
  "includeGraphics": true,
  "includeStatistics": true,
  "includeDiagnosticSummary": true,
  "timeRange": {
    "startTime": "2026-01-15T14:30:00Z",
    "endTime": "2026-01-17T14:30:00Z"
  }
}'

```

Response (202 Accepted):

```
{
  "success": true,
  "data": {
    "reportId": "report_507f1f77bcf86cd799439014",
    "status": "processing",
    "estimatedCompletionTime": "30s",
    "jobId": "job_507f1f77bcf86cd799439015"
  }
}
```

Obtener Reporte Generado

Endpoint: GET /reports/{reportId}

```
curl -X GET https://api.horizon-medical.com/v2/reports/re-
port_507f1f77bcf86cd799439014 \
-H "Authorization: Bearer {accessToken}"
```

Response (cuando está listo):

```
{
  "success": true,
  "data": {
    "id": "report_507f1f77bcf86cd799439014",
    "sessionId": "session_507f1f77bcf86cd799439013",
    "status": "completed",
    "format": "pdf",
    "downloadUrl": "https://storage.horizon-medical.com/reports/re-
port_507f1f77bcf86cd799439014.pdf",
    "createdAt": "2026-01-15T14:35:00Z",
    "expiresAt": "2026-02-14T14:35:00Z",
    "fileSize": 2458624,
    "insights": {
      "summaryFindings": "Ritmo sinusal normal, 47 ESVs aisladas",
      "criticalFindings": [],
      "recommendations": ["Seguimiento en 4 semanas"]
    }
  }
}
```

Descargar Reporte

Endpoint: GET /reports/{reportId}/download

```
curl -X GET https://api.horizon-medical.com/v2/reports/re-
port_507f1f77bcf86cd799439014/download \
-H "Authorization: Bearer {accessToken}" \
--output report.pdf
```

Alertas

Listar Alertas

Endpoint: GET /alerts

```
curl -X GET "https://api.horizon-medical.com/v2/alerts?severity=critical&status=unread&limit=50" \
-H "Authorization: Bearer {accessToken}"
```

Parámetros:

```
severity: string (critical, high, medium, low, info)
status: string (unread, read, acknowledged)
patientId: string (opcional)
since: ISO8601 datetime
sortBy: string (timestamp, severity)
```

Obtener Alerta

Endpoint: GET /alerts/{alertId}

```
curl -X GET https://api.horizon-medical.com/v2/alerts/alert_507f1f77bcf86cd799439016 \
-H "Authorization: Bearer {accessToken}"
```

Response:

```
{
  "success": true,
  "data": {
    "id": "alert_507f1f77bcf86cd799439016",
    "sessionId": "session_507f1f77bcf86cd799439013",
    "patientId": "patient_507f1f77bcf86cd799439012",
    "type": "arrhythmia",
    "subtype": "atrial_fibrillation",
    "severity": "critical",
    "status": "unread",
    "timestamp": "2026-01-15T16:45:00Z",
    "message": "Atrial fibrillation detected",
    "details": {
      "heartRate": 145,
      "duration": 600,
      "confidence": 0.99
    },
    "actions": [
      {"type": "contact_patient", "status": "pending"},
      {"type": "contact_physician", "status": "pending"}
    ]
  }
}
```

Marcar Alerta como Revisada

Endpoint: PATCH /alerts/{alertId}/acknowledge

```
curl -X PATCH https://api.horizon-medical.com/v2/alerts/
alert_507f1f77bcf86cd799439016/acknowledge \
-H "Authorization: Bearer {accessToken}" \
-H "Content-Type: application/json" \
-d '{
  "notes": "Paciente evaluado, sin síntomas"
}'
```

WebSocket Events

Conexión WebSocket

```
const socket = io('wss://api.horizon-medical.com', {
  auth: {
    token: 'eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...'
  }
});

// Conectado
socket.on('connect', () => {
  console.log('Conectado al servidor de eventos');
  socket.emit('subscribe_session', {
    sessionId: 'session_507f1f77bcf86cd799439013'
  });
});
```

Eventos en Tiempo Real

Nuevo Dato ECG

```
socket.on('ecg_data', (data) => {
  console.log('Nuevo ECG:', {
    timestamp: data.timestamp,
    heartRate: data.heartRate,
    derivation: data.derivation,
    samples: data.samples
  });
});
```

Alerta Crítica

```
socket.on('alert', (alert) => {
  console.log('¡ALERTA!', {
    type: alert.type,
    severity: alert.severity,
    message: alert.message
  });
});
```

Cambio de Estado

```
socket.on('session_status_changed', (data) => {
  console.log('Estado de sesión:', {
    sessionId: data.sessionId,
    status: data.status, // active, paused, completed
    timestamp: data.timestamp
  });
});
```

Sincronización Completada

```
socket.on('sync_completed', (data) => {
  console.log('Sincronización:', {
    sessionId: data.sessionId,
    dataPoints: data.dataPointsReceived,
    timestamp: data.timestamp
  });
});
```

Manejo de Errores

Tipos de Error

```
{
  "success": false,
  "error": {
    "code": "ERROR_CODE",
    "message": "Mensaje legible para usuario",
    "details": {}
  }
}
```

Códigos de Error Comunes

AUTHENTICATION_REQUIRED	→ 401 Unauthorized
INVALID_CREDENTIALS	→ 401 Unauthorized
TOKEN_EXPIRED	→ 401 Unauthorized
INSUFFICIENT_PERMISSIONS	→ 403 Forbidden
RESOURCE_NOT_FOUND	→ 404 Not Found
VALIDATION_ERROR	→ 422 Unprocessable Entity
RATE_LIMIT_EXCEEDED	→ 429 Too Many Requests
DATABASE_ERROR	→ 500 Internal Server Error
SERVICE_UNAVAILABLE	→ 503 Service Unavailable
INVALID_FILE_FORMAT	→ 400 Bad Request
DEVICE_NOT_CONNECTED	→ 400 Bad Request
SESSION_NOT_ACTIVE	→ 400 Bad Request
INSUFFICIENT_DATA	→ 400 Bad Request

Ejemplo de Manejo de Error

```
try {
  const response = await fetch(
    'https://api.horizon-medical.com/v2/patients',
    {
      headers: {
        'Authorization': `Bearer ${token}`,
        'Content-Type': 'application/json'
      }
    }
  );

  if (!response.ok) {
    const errorData = await response.json();

    switch (response.status) {
      case 401:
        // Refresh token o redirigir a login
        break;
      case 429:
        // Esperar según X-RateLimit-Reset
        break;
      case 500:
        // Reintentar con backoff exponencial
        break;
    }
  }

  const data = await response.json();
  return data.data;
} catch (error) {
  console.error('API Error:', error);
}
```

Ejemplos de Integración

Python

```
import requests
import jwt
from datetime import datetime, timedelta

class HorizonMedicalAPI:
    def __init__(self, api_key, api_secret):
        self.base_url = "https://api.horizon-medical.com/v2"
        self.api_key = api_key
        self.api_secret = api_secret
        self.token = None
        self.token_expires_at = None

    def get_token(self):
        if self.token and datetime.now() < self.token_expires_at:
            return self.token

        payload = {
            'iss': self.api_key,
            'iat': datetime.utcnow(),
            'exp': datetime.utcnow() + timedelta(hours=1)
        }

        token = jwt.encode(
            payload,
            self.api_secret,
            algorithm='HS256'
        )

        self.token = token
        self.token_expires_at = datetime.now() + timedelta(hours=1)
        return token

    def get_patients(self, page=1, limit=20):
        headers = {
            'Authorization': f'Bearer {self.get_token()}',
            'Content-Type': 'application/json'
        }

        response = requests.get(
            f'{self.base_url}/patients',
            headers=headers,
            params={'page': page, 'limit': limit}
        )

        response.raise_for_status()
        return response.json()['data']

# Uso
api = HorizonMedicalAPI('your_api_key', 'your_api_secret')
patients = api.get_patients(page=1, limit=50)
```

JavaScript/TypeScript

```
class HorizonMedicalAPI {
  private baseUrl = 'https://api.horizon-medical.com/v2';
  private accessToken: string | null = null;
  private refreshToken: string | null = null;

  async login(email: string, password: string): Promise<void> {
    const response = await fetch(`${this.baseUrl}/auth/login`, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ email, password })
    });

    if (!response.ok) throw new Error('Login failed');

    const { data } = await response.json();
    this.accessToken = data.accessToken;
    this.refreshToken = data.refreshToken;
  }

  async getPatients(page: number = 1, limit: number = 20) {
    const response = await fetch(
      `${this.baseUrl}/patients?page=${page}&limit=${limit}`,
      {
        headers: {
          'Authorization': `Bearer ${this.accessToken}`
        }
      }
    );

    if (response.status === 401) {
      await this.refreshAccessToken();
      return this.getPatients(page, limit);
    }

    const { data } = await response.json();
    return data;
  }

  private async refreshAccessToken(): Promise<void> {
    const response = await fetch(`${this.baseUrl}/auth/refresh`, {
      method: 'POST',
      headers: {
        'Authorization': `Bearer ${this.refreshToken}`
      }
    });

    const { data } = await response.json();
    this.accessToken = data.accessToken;
  }
}
```

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Para soporte API: api-support@horizon-medical.com

Documentación interactiva: <https://api.horizon-medical.com/docs>